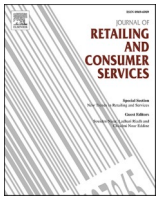
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Decoding influencer marketing effectiveness on instagram: Insights from image, text, and influencer features

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ABSTRACT

Influencer marketing has become a crucial strategy for modern brands. By collaborating with influencers to publish sponsored posts on social media platforms, brands can leverage influencer popularity and follower engagement to enhance brand exposure and attract potential consumers. This study aims to predict the popularity of sponsored posts on Instagram and analyze the factors influencing their effectiveness. To achieve this, four distinct feature sets were extracted from sponsored posts: image visual features, image topic features, text topic features, and influencer features. These features were used individually and in combination as predictive variables to develop models for predicting post popularity using various methods. Experimental results demonstrate that the predictive models achieve strong performance, with the best results obtained when incorporating all four feature sets, highlighting the importance of considering multiple factors in evaluating sponsored post effectiveness. Furthermore, this study employs the Taguchi experiments to analyze the relative contribution of the four feature sets to the post popularity and utilizes odds ratio analysis from logistic regression to provide detailed insights into the impact of individual features. By examining the influence of visual, textual, and influencer-related factors, this study offers valuable guidance for brands in selecting influencers and optimizing post content, providing deeper insights into influencer marketing strategies.

1. Introduction

Influencer marketing refers to a marketing strategy in which brands collaborate with popular social media users to promote brands' products or services (Mckinsey, 2023). Influencers are content creators on social media with expertise in specific fields who own sizable social media followers and can influence their followers' behavior as consumers. For brands, Influencers hold significant marketing value (Lou and Yuan, 2019; Mckinsey, 2023). The origins of influencer marketing can be traced back to the early 2000s, when blogging emerged as a popular form of online content creation and became increasingly widespread. With the rapid development of Web 2.0 in the mid-2000s, interactive, social, and user-generated content (UGC) websites and platforms quickly gained momentum, allowing users to create content within virtual communities. As platforms like Facebook, Twitter, Instagram, and YouTube emerged, they enabled influencers to gain popularity and prompted brands to collaborate with influencers to promote their products or services, ultimately giving rise to influencer marketing (Wilkens, 2023). An analysis of over 2000 influencer marketing posts in 2022 indicated that influencer marketing strategies do deliver positive returns on investment for brands (Mckinsey, 2023). This success has drawn significant attention and investment from many brands and advertising agencies. Influencer Marketing Hub (2025) indicated that influencer marketing is projected to reach $32.55 billion in global market size by 2025, compared to $24 billion in 2024 and a modest $1.4 billion in 2014. Over 80 % of marketers affirm influencer marketing as a highly effective strategy.

Instagram was launched in 2010, quickly rising to prominence among numerous social media platforms due to its minimalist design and emphasis on photo sharing. This appeal resonated strongly with younger generations, making Instagram one of today's most popular social media platforms (Dhanesh and Duthler, 2019). The GlobalWebIndex (GWI) 2024 survey reveals that Instagram is now the world's favorite social media platform, with 16.5 percent of internet users aged 16 to 64 selecting it as their preferred platform. Meanwhile, according to the 2023 Taiwan Top 100 Influencers Insight Report from the KOL Radar platform (https://www.kolradar.com/), over 50,000 brands in Taiwan are using Instagram to find optimal partners, with 90 % of the top 100

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influencers actively engaging on the platform. These statistics demonstrate the substantial global influence of Instagram and its popularity in Taiwan as well. For brands, Instagram serves as a platform where they can gather consumers, foster connections, and encourage social interactions between both parties (Jin et al., 2019). Consequently, when brands consider where to share their marketing information, Instagram has become the most popular choice (Hur et al., 2020). Furthermore, in a 2022 survey of more than 2000 global brands, about 80 % reported using Instagram for influencer marketing (Mckinsey, 2023). Research also shows that visuals are more attention-grabbing than text content (Chan et al., 2023), and the rich image-based content on Instagram supports influencer marketing more effectively than text-based platforms like Twitter (Haenlein et al., 2020).

On the highly popular platform Instagram, for brands to achieve successful influencer marketing, they must not only focus on the influencer's social media engagement, characteristics, and popularity to positively impact brand product sales but also ensure that the marketing post content captures consumers' attention. Research on the success of influencer marketing has primarily explored two main factors: influencers' personal traits and the post content, including images and texts (Vrontis et al., 2021). In terms of influencer traits, studies have examined factors such as influencers' styles, credibility, attractiveness, popularity, and alignment with brand image, analyzing their influence on marketing performance (e.g., Arora et al., 2019; Ye et al., 2021). Ki and Kim (2019) pointed out that when influencers are visually appealing and convey an image of expertise, consumers are more likely to develop a positive attitude toward the influencer's marketing activities, thereby prompting their purchase intentions and behaviors. Regarding post content, research has focused on elements such as text length, word count, image aesthetics, and color tone, as well as the uniqueness and originality of posts (e.g., Ki and Kim, 2019; Purba et al., 2021; Vrontis et al., 2021; Mahdikhani, 2022). Textual and visual elements serve as communication mediums between influencers and their followers, helping maintain their relationships (Feng et al., 2021). The originality and uniqueness of content can also enhance consumers' willingness to interact with influencers and increase their trust in influencer recommendations (Vrontis et al., 2021). Particularly on Instagram, the visual appeal of posts is a key factor. Therefore, brands should emphasize both the quality of images and texts when designing marketing posts, ensuring they are rich in informational value and appeal to boost brand awareness and purchase intentions (Lou and Yuan, 2019).

However, past research on extracting post features and examining their effects on the post popularity has some limitations for influencer marketing. First, most studies do not specifically focus on sponsored posts of influencer marketing. As a result, the posts analyzed may come from non-influencer general users, and the content may be unrelated to product sales. Second, most studies do not simultaneously consider the features of post creator, post images, and post texts, which holistically influence the success of a post. Additionally, the features extracted may lack semantic or practical interpretability, or the research may focus solely on improving post popularity predictions without exploring how these features exert their influence, thereby limiting the applicability of the analysis results. For example, Wang et al. (2023) studied the popularity of Flickr posts, whereas overlooked features related to post creators. Mahdikhani (2022) focused solely on the textual content of Twitter, using various methods to characterize content, such as TF-IDF, Doc2Vec, and topic modeling for post popularity prediction. However, it lacks an explanation of how these features affect the popularity. Purba et al. (2021) studied Instagram posts and emphasized the analysis of post images and post creators but lacked consideration of the semantic features of textual content. Moreover, while the study has quantified the aesthetic and technical qualities of images as image features, they struggle to explain which image elements contribute to aesthetic or technical quality, thereby enhancing post popularity.

In this highly competitive digital era, selecting the right influencers, creating eye-catching branded content, and maximizing the value of influencer marketing budgets have become both challenges and opportunities for brand management. Successful influencer marketing can not only foster more positive and trusting attitudes toward a brand but also increase engagement on social media, thereby enhancing brand awareness and influence among consumers. Additionally, it can evoke feelings of envy, sparking consumers' desire and pursuit of the brand (Jin et al., 2019). Undoubtedly, all of these rely on prior understanding and mastery of the key factors of success in influencer marketing. This study explored the key factors influencing the success of sponsored posts in influencer marketing on Instagram within the Taiwan market. Specifically, it examined influencer features such as gender, field of expertise, Instagram-specific metrics (e.g., follower count and post count), and metrics on other social media platforms. To characterize post content, text data were featured as probabilistic topics using topic modeling techniques. For images, using embedding techniques to obtain overall visual summary features is a common approach in current image analysis applications. However, while embedded features can effectively represent the original image, they lack semantic meaning. Therefore, in addition to extracting visual features, this study converts images into textual captions, from which semantic topics are extracted for further content interpretation. Using influencer features, text topic features, image visual features, and image topic features, predictive and analytical models were developed to predict post popularity. The impact of these features was further analyzed using Taguchi experiments and odds ratios.

By interpreting these semantically meaningful features and their influence on sponsored content, this study provided insights into influencer marketing on Instagram. The findings serve as a valuable reference for brands and influencers to optimize collaboration strategies and enhance engagement effectiveness.

2. Literature review

This study attempted to identify and understand the influential features of the popularity of sponsored posts published by influencers on Instagram by building post popularity prediction models and analyzing the predictor effects. This section will review relevant literature from four perspectives: the influence of social media influencers, influencer marketing, user-generated content, and post popularity prediction and its applications.

2.1. The influence of social media influencers

Social media influencers are individuals who are highly active on various social platforms, frequently engaging in discussions within specific fields—such as fashion, lifestyle, photography, and travel—while providing the latest information on these topics (Loeper et al., 2014). Gaining fame through their social media presence, these influencers are often creative and regularly post engaging and appealing content, cultivating a large and loyal follower base (De Veirman et al., 2019). They post across multiple social platforms, with Instagram being the most frequently used for influencer marketing (Zote, 2024).

Influencers often act as opinion leaders or experts within their fields and have the ability to subtly shape their followers' thoughts, attitudes, and behaviors (Dhanesh, 2019). When influencers recommend products or services that interest their followers, these recommendations are perceived as friend-like advice (Kaabachi et al., 2021), making them highly effective; about 60 % of respondents report purchasing products or services based on influencer recommendations (Sharma, 2023). Thus, followers often rely on influencers' opinions when making purchase decisions (Jin et al., 2019). Through influencers' social media accounts, followers gain insights into influencers' lifestyles, personal experiences, and viewpoints (De Veirman et al., 2017). Users on social media tend to follow influencers who share common interests, personality traits, and lifestyles. This creates a sense of mutual understanding (Sharma, 2023) which enhances emotional resonance and fosters a sense of connection

between followers and influencers.

Many studies compared influencers to traditional celebrities, suggesting that influencers engage more with their followers and wield greater influence, particularly within niche communities (Masuda et al., 2022). Influential influencers are also more effective in generating positive brand attitudes among consumers (Jin et al., 2019). Consumers are more likely to relate to and trust influential influencers because they feel a greater sense of similarity with them (Schouten et al., 2021). As a result, brands increasingly leverage the relationship and influence between influencers and their followers to promote their products or services through influencer marketing (Dhanesh and Duthler, 2019), driving the ongoing development of influencer marketing strategies on social media, especially on Instagram (Belanche et al., 2021).

Numerous studies have proposed various metrics for assessing influencer impact. Arora et al. (2019) considered follower count, likes, comments, shares, and sentiment scores across Instagram, Facebook, and Twitter pages and posts to propose six metrics: overall footprint, engagements & outreach, hourly engagement velocity, daily engagement velocity, audience sentiment, and posting rate, to evaluate influencers' influence. Muñoz et al. (2022) focusing on Twitter analyzed engagement based on tweet volume and follower count to rank influencers. These metrics and methodologies provide valuable frameworks for evaluating influencer impact, helping to understand influencers' performance and influence on social media. This study will also collect influencers' features—such as post count, follower count, and engagement rate—as predictors of post popularity. By analyzing these data, brands can gain better insights into influencer impact and audience engagement, aiding in the selection of suitable influencers for collaboration to maximize brand exposure and influence on social media, ultimately enhancing the effectiveness of influencer marketing.

2.2. Influencer marketing

Influencer marketing refers to a strategy in which brands and advertisers provide material support to influencers, enabling them to post on social media to promote products, services, brands, ideas, or viewpoints (Campbell and Farrell, 2020). Influencers often post photos or videos to engage followers, allowing them to experience the influencer's lifestyle virtually, which in turn helps promote products (Feng et al., 2021).

Influencer marketing is active across various social platforms. Research has examined eight platforms—Facebook, Instagram, YouTube, Twitter, Snapchat, TikTok, LinkedIn, and Pinterest—to understand how each influences user satisfaction with interactions with influencers. Among these, Instagram offers features like stories, shopping tools, and live streaming, which make users find interactions with influencers on Instagram enjoyable (Lou et al., 2023). To date, Instagram influencers have been particularly popular among brands, as the personal brand equity they create provides high returns on investment. According to the Influencer Marketing Hub report, successful influencer marketing campaigns yield $18 in media value for every dollar spent. Instagram influencers are highly effective at disseminating information about new products, spreading trends, and driving sales (Jin et al., 2019). Making partnerships with influential Instagram influencers is a critical success factor in social media marketing strategies (De Veirman et al., 2017).

Numerous studies have confirmed the benefits of influencer marketing for brands. Brands can leverage influencers' status, credibility, and popularity to build their image among the influencer's large follower base (De Veirman et al., 2017). A higher follower count on Instagram positively affects followers' attitudes toward influencers (De Veirman et al., 2017). As a result, most studies on influencers' post popularity have considered follower count as a critical feature. The expertise, fame, and positive reputation of influencers collectively create the potential to attract a large audience. In addition, the alignment between influencers' fields of expertise and the marketed products is also crucial. While it may not significantly impact the influencers' credibility or professionalism, it directly influences consumers' attitudes toward the advertisement and their purchase intentions (Ye et al., 2021). Therefore, brands should consider the compatibility of influencers' fields with their own brand image, indicating another impactful characteristic worth exploring.

Moreover, an analysis of Instagram post engagement revealed that sponsored posts typically achieve higher engagement compared to non-sponsored posts. This is attributed to influencers investing more time and effort in creating sponsored content, and the high hedonic value embedded in sponsored posts may be a potential factor driving higher engagement (Argyris et al., 2020). Additionally, as influencers' follower count increases, post engagement also tends to rise, further enhancing brand engagement on social media through these sponsored posts. Furthermore, a study proposed an integrated model—the Social Media Influencer Value Model—and found through survey analysis that the informational value of influencer-generated content positively affects followers' trust in sponsored posts, which in turn influences brand awareness and purchase intentions (Lou and Yuan, 2019). These findings collectively underscore that post content is a critical factor influencing user attitudes and engagement levels.

This study focused on Instagram influencers as the research subjects, taking into account their personal features, such as follower count and recognized fields of expertise, while also analyzing the informational content of their posts to explore the factors influencing the effectiveness of sponsored posts. Successful influencer marketing enables brands not only to achieve greater interaction on social media platforms but also to enhance brand awareness and further stimulate consumer purchase actions. This, in turn, helps brands build stronger market competitiveness and foster greater consumer loyalty.

2.3. User-generated content

UGC includes text, images, audio, and videos posted by users on social media, e-commerce sites, video-sharing platforms, and review websites (Wu et al., 2024). Social media has become a popular channel for content creators to share experiences and trending information across areas such as fashion, health, hobbies, and daily life (Zhuang et al., 2023). Many studies explored how UGC on social media platforms can benefit brands. Sreejesh et al. (2020) suggested that product-related UGC aids consumers in understanding products, forming perceptions, and fostering purchase intentions. Wang et al. (2021) indicated that more and more businesses are leveraging UGC as a content marketing strategy because of its ability to build consumers' brand awareness and loyalty. McClure and Seock (2020) highlighted that high-quality UGC, such as well-crafted testimonials has been shown to enhance credibility and attractiveness, which leads to greater consumer engagement with brand social media. The study also indicated that authentic user experiences are often more trustworthy and resonate more than direct brand promotion. Consequently, brands frequently collaborate with influential creators to share sponsored posts that reflect their personal experiences, which resonate with followers. As different platforms offer varied engagement opportunities, Roma and Aloini (2019) analyzed how brand-related UGC characteristics vary across social media platforms, Facebook, Twitter, and YouTube. Findings indicate that Facebook and Twitter facilitate greater user engagement in comparison to YouTube, especially on the UGC characteristics related to the real-time and ubiquitous sharing of increasingly richer content. Instagram, favored by younger generations, is recommended as a strategic platform for marketers aiming to boost consumer engagement (Djafarova and Bowes, 2021).

In UGC types, text and images are the most common forms and have become the primary focus of numerous studies and analyses. Text-based content analysis often involves sentiment evaluations and topic modeling to gauge user attitudes and opinions and summarize user concerns. Pathak et al. (2021) applied these methods to help brands in

market research and product improvements. Jiang et al. (2024) identified distinct topics from China's news posted on social media by using BERTopic technique and associated the topics with news engagement metrics (e.g., number of views, likes, and comments) in order to provide insights into content dissemination. Kim et al. (2022) explored consumers' preferences and perceptions of genderless fashion trends from consumer-posted text on blogs and online communities through text mining and latent Dirichlet allocation (LDA)-based topic modeling analysis. On social media, images may have a more significant impact than text, as visuals tend to attract attention quickly and are more memorable (Li and Xie, 2020). Li and Xie (2020) used data sets of social media posts about major airlines and sport utility vehicle brands collected from Twitter and Instagram to examine the influence of image content on social media engagement. Yu and Egger (2021) uncovered the relationship between color and user engagement based on pictures with different features by classifying tourism photos on Instagram using machine learning. Abell and Biswas (2023) investigated how the type of food (healthy vs. unhealthy) featured next to a person in a social media image ad or influencer image post impacts online engagement.

These studies demonstrated the importance of analyzing both textual and visual features in understanding user opinions and engagement. Therefore, our study, aimed at identifying key success factors in influencer marketing, will also consider how the textual and visual characteristics of influencers' sponsored content impact engagement metrics, such as the count of likes. This approach integrates insights from previous research into our analysis, enabling a more comprehensive understanding of the factors driving user interactions with influencer posts.

2.4. Post popularity prediction and its applications

In the era of social media, collaborations between brands and influencers have become a prevalent marketing strategy, as brands aim to capture consumer attention through sponsored posts (Antoniadis et al., 2019). As a result, measuring the effectiveness of sponsored posts following such collaborations has become one of the primary objectives of marketing campaigns, as this assessment informs future partnerships. However, the effectiveness of a post is influenced by numerous factors, including the influencer's background, the content of the post, and its format. Accurately predicting the popularity of a post, exploring the factors critical to affecting popularity, and identifying optimal collaboration strategies is thus a compelling research problem worthy of deeper investigation.

In recent years, a growing body of literature has emerged on the prediction of social media post popularity. These studies span various platforms, such as Instagram, Twitter, Facebook, and Flickr, and explore a diverse range of features, including follower count, textual content, images, post types, and post attributes (e.g., posting time, geotags). Moreover, a variety of research methodologies have been employed, including statistical regression, machine learning algorithms, and deep learning models.

Purba et al. (2021) extracted features from images, hashtags, and user history to predict the engagement rate of Instagram posts. The features of images included aesthetic and technical qualities, while hashtags were assessed based on their quantity and popularity. User history considered features such as the number of followings, the number of posts, and biography length. The research compared several regression techniques, and it was found that image quality, posting time, and type of image highly impact engagement rate. The prediction accuracy reached up to 73.1 % using the support vector regression. Carta et al. (2020) applied decision tree and XGBoost classification models to predict the popularity level of Instagram posts. The features considered included the number of words, the number of users tagged, the number of chosen hashtags, sentiment scores, and the presence of emojis in the caption. The results indicated that XGBoost demonstrated superior accuracy. Moreover, they found that if a post's number of likes exceeded the average number of likes for the most recent 50 posts, the post was likely to be highly popular. Zohourian et al. (2018) used some image and video context features to predict the number of likes a post obtains as a meaning of popularity through some regression and classification methods. Their study considered time features, text features such as captions and the number of hashtags, common features such as the number of comments, follower count, and post type, and RGB values for images and videos. Notably, for videos, the exclusive features of music, narration, and cover were considered. Through the experiments, the study got the results of popularity score prediction with 0.002 in RMSE and popularity class prediction with 90.77 % accuracy.

In addition to Instagram, numerous studies focus on other social media platforms. For instance, Wang et al. (2023) utilized datasets from Flickr to predict the popularity scores of posts using features of image, image attribute, text, and text attribute. The image features were extracted with ResNet and the image attribute features were generated using an image attribute predictor constructed by image captions and GloVe embeddings. The Bi-LSTM model was used to obtain text features, and the text attribute features such as data, tags count, and location were added through data preprocessing. Then, these features were reconstructed with a multimodal hierarchical fusion model and merged into a feature vector for popularity prediction. Their findings revealed that XGBoost demonstrated the best predictive performance, and both image and textual features contributed significantly to the prediction. Antoniadis et al. (2019) conducted a study on Facebook, employing regression analysis to identify the impact of various features, such as post type (image or video), sentiment of texts, post length, and publication date. The results showed that images and the sentiment of texts had a significant positive influence on all popularity metrics. Post length also had a positive but smaller effect, while the publication date did not show a significant impact. Mahdikhani (2022) focused on Twitter, predicting the popularity of tweets during the Covid-19 pandemic. They applied LDA to analyze the topics within tweet texts and vectorized the texts using TF-IDF and Doc2vec. Popularity was defined based on whether the number of shares exceeded one. The results indicated that, compared to other content feature sets, the combination of topic vectors and TF-IDF features achieved the highest accuracy, with the ensemble voting method performing best among the models. Similarly focusing on Twitter, Daga et al. (2020) extracted features from textual content using TF-IDF and Doc2Vec while simultaneously considered post attributes, including the number of likes, shares, and posting time for post popularity predictions. Five machine learning models—Logistic Regression, SVM, Random Forest, Neural Network, and Bayesian Classifier—were employed to predict the number of likes and shares, categorized into four classes. The results demonstrated that TF-IDF consistently outperformed Doc2Vec across all tests. Although Doc2Vec is capable of capturing contextual information, the findings suggest that the typically short nature of post texts makes them better suited for TF-IDF.

Synthesizing insights from the aforementioned literature, it is evident that a post creator's personal attributes, along with post text and image content, play significant roles in predicting post popularity. Textual and visual features are complementary, both contributing to enhanced prediction accuracy (Mazloom et al., 2016; Wang et al., 2023). However, existing research often limits itself to vectorizing image and textual features, making it difficult to interpret their specific impacts. For images, even when features such as RGB values, brightness, aesthetic quality, and style are extracted, it remains challenging to pinpoint the elements that influence post popularity. Similarly, while textual content can be analyzed through topics or keywords to identify its relationship with popularity, few studies provide deeper explanations. For studies that convert post text and images into embedding vectors as features using, for example, Doc2Vec and ResNet, while they can improve prediction performance, they fail to explain the semantic meaning of these features. Furthermore, post popularity is not solely determined by content but also by the personal attributes of the post creators. Although many studies have included factors like follower count and the number of posts, their significance often remains

inconclusive.

This study aimed to further investigate the image features of sponsored posts in influencer marketing. Beyond extracting visual features, it attempted to convert image content into semantic topics to improve interpretability. For textual content, topic modeling will be employed to understand the topics within the text and analyze their influence on marketing effectiveness. Regarding influencer features, the study will collect data such as gender, category, number of posts, follower count, and social engagement metrics from other platforms as potential factors influencing the popularity of sponsored posts. By integrating these multiple features, this research seeks to analyze their combined impact on post popularity. The ultimate goal is to provide brands with deeper insights into the factors driving the popularity of influencer posts, thereby enhancing prediction accuracy and informing marketing strategies.

3. Methodology

This study analyzed sponsored posts published by influencers on Instagram, aiming to extract various types of features to predict post popularity which were defined as two levels based on the number of likes. Furthermore, the impact of the features was analyzed to help brands better understand the effectiveness of influencer marketing. The overall analysis framework is depicted in Fig. 1.

First, influencer data is collected, including influencer features and their Instagram accounts. Using Python with the Instagram Graph API, data on influencer posts, such as images, text, and the number of likes, was gathered. Sponsored posts were identified, and their images were processed using ResNet50 to extract visual features. Images were also converted into textual descriptions, followed by topic analysis to derive image topic features. The textual content of sponsored posts underwent topic analysis to extract textual topic features.

Subsequently, influencer features, image visual features, image topic features, and textual topic features were used to build predictive models utilizing various machine learning models, deep neural network models, and voting-based ensemble models. Furthermore, the study employed the Taguchi experiments to evaluate the importance of different types of features in predicting the popularity of sponsored posts. The influence of

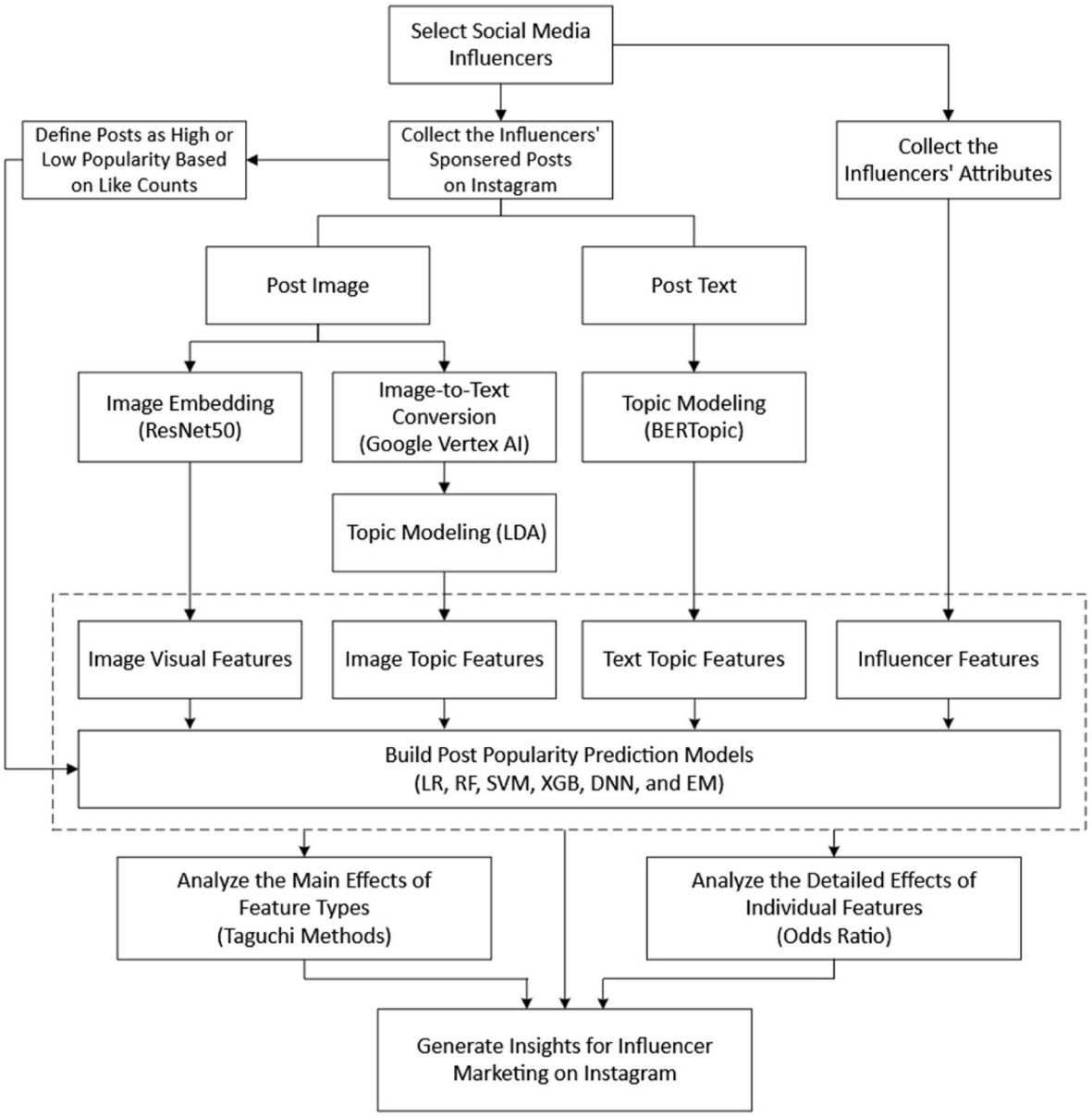


Fig. 1. Analysis process.

detailed features within each feature type was then interpreted using the odds ratio results of logistic regression on the model's predictive power.

Finally, the results are discussed to provide brands with deeper insights and more precise strategies for influencer marketing.

3.1. Influencers and their features

This study focused on the Taiwanese market and the influencers included in the study and some of their feature data were referenced from the "2023 Taiwan Top 100 Influencer Insight Report" jointly released by Taiwan tech media outlets "TechOrange" and "KOL Radar" on November 1, 2023 (TechOrange & KOL Radar, 2023). The report investigated over 250,000 user accounts in Taiwan between June 1, 2022 and July 31, 2023 across Facebook, Instagram, and YouTube, which are the three most popular social media platforms in Taiwan. The top 100 Influencer accounts were finally selected and reported according to their social media data and influence evaluation. Since this study focused on influencer marketing on Instagram as the primary platform, the top influencers without Instagram accounts or who belong to meme accounts were excluded, leaving 67 remaining.

Table 1 shows the influencer features considered in this study. For the 67 influencers, we used their Instagram IDs and Instagram Graph API to collect their post counts and following counts on Instagram. Other features, including follower counts, engagement rates, and view counts across the three platforms, were collected from the report. However, since not all influencers are active across multiple platforms, some data were missing. Therefore, this study used Instagram follower counts as the primary reference, grouped in increments of 50,000. For missing data on other platforms, the average value within the corresponding group is used for imputation. For example, if an influencer has 135,000 Instagram followers but no data on other platforms, the average data values from influencers in the 100,000–150,000 Instagram follower range are used to fill in the missing data. Additionally, the KOL Radar website (https://www.kolradar.com) provides information on influencers' fields of expertise and genders, which are categorical features. There are a total of 10 fields: lifestyle, fashion, sports, travel, affection, technology, education, entertainment, beauty, and food. An influencer can be associated with multiple fields based on their past active engagement on social media. Gender was divided into three categories: male, female, and non-gendered. These categorical features were binarized for subsequent analyses.

Table 1
Influencer features.

No.	Feature	type	No.	Feature	type
1	Post Counts on Instagram	continuous	13	Fashion	binary
2	Following Counts on Instagram	continuous	14	Sports	binary
3	Follower Counts on Instagram	continuous	15	Travel	binary
4	Engagement Rates on Instagram	continuous	16	Affection	binary
5	View Counts on Instagram	continuous	17	Technology	binary
6	Follower Counts on Facebook	continuous	18	Education	binary
7	Engagement Rates on Facebook	continuous	19	Entertainment	binary
8	View Counts on Facebook	continuous	20	Beauty	binary
9	Suscriber Counts on YouTube	continuous	21	Food	binary
10	Engagement Rates on YouTube	continuous	22	Male	binary
11	View Counts on YouTube	continuous	23	Female	binary
12	Lifestyle	binary	24	Non-Gendered	binary

3.2. Sponsored post collection and post popularity definition

Instagram posts (an example is shown in Fig. 2) published between June 1, 2022 and July 31, 2023 from 67 influencers were retrieved using the Instagram Graph API. A total of 19,082 posts were collected, with each post including its image (or video thumbnail), text content, and the number of likes. Observations revealed that sponsored posts on Instagram frequently use hashtags to tag brands or products, enhancing the reach and engagement of marketing campaigns (Kim and Phua, 2020). Therefore, this study retained only posts containing hashtags, identifying them as sponsored content for analysis.

The sponsored posts were then sorted based on the number of likes. Due to significant variability in like counts, as illustrated in Fig. 3, the top and bottom 10 % of posts by likes were excluded to reduce the influence of outliers. The remaining posts were further categorized into high-popularity and low-popularity groups, defined as the top and bottom 30 % of posts by likes, respectively. This filtering process resulted in a final dataset of 4690 sponsored posts, with 2345 posts in both high- and low-popularity categories. The images and texts from these posts then underwent further feature extraction for analysis.

3.3. Image feature extraction

Two types of features were extracted from the image of a sponsored post: visual features and topic features. For visual features, a transfer deep learning model was employed to generate image embeddings as feature representations. However, such image-embedding features are less semantically interpretable. Therefore, image topic features were extracted by converting images into textual captions for further topic analysis. In addition to considering the holistic visual presentation, it enables an understanding of the specific elements and topics within the images, enhancing interpretability. These two types of features allowed for a more comprehensive examination of how images influence post popularity.

3.3.1. Image visual feature extraction

This study employed the ResNet50 (He et al., 2016) to extract visual features from post images. ResNet50 is a deep convolutional neural network with 50 layers that utilizes residual blocks to effectively address the vanishing gradient problem. By enabling the training of very deep models with fewer parameters, ResNet50 achieves high accuracy and robustness. Pretrained on large-scale datasets, ImageNet (Deng et al., 2009), ResNet50 is widely applied in tasks like image classification, object detection, and segmentation, making it a popular choice in computer vision research and applications. For example, Li and Lima (2021) successfully used ResNet50 for facial emotion recognition. Gujjar et al. (2021) evaluated five pre-trained image models, including VGG16, VGG19, ResNet50, MobileNet, and MobileNetV2, for image classification and prediction, with ResNet50 achieving the highest accuracy. Furthermore, the Keras, an open-source library for deep learning, compared various models in terms of size, parameter count, computation time, and classification accuracy on the ImageNet dataset (https://keras.io/api/applications), highlighting the outstanding performance of ResNet50.

This study utilized the ResNet50 pre-trained model from the Keras library in Python as the base model for analysis, with additional layers tailored to the specific task. A Global Average Pooling layer was added to compress the feature maps into a fixed-size vector, followed by a dense layer with ReLU activation to learn compact and abstract feature representations. Finally, a custom 2-dimensional output layer with Softmax activation was introduced to align with the high-popularity and low-popularity classification requirements of our prediction task. All input images were standardized to a fixed size of 224×224 by adopting proportional resizing and zero padding. Data augmentations, including image rotation, translation, cropping, scaling, shifting, and horizontal flipping, were employed to enhance the diversity of the training dataset.



Fig. 2. An example of sponsored post on instagram.

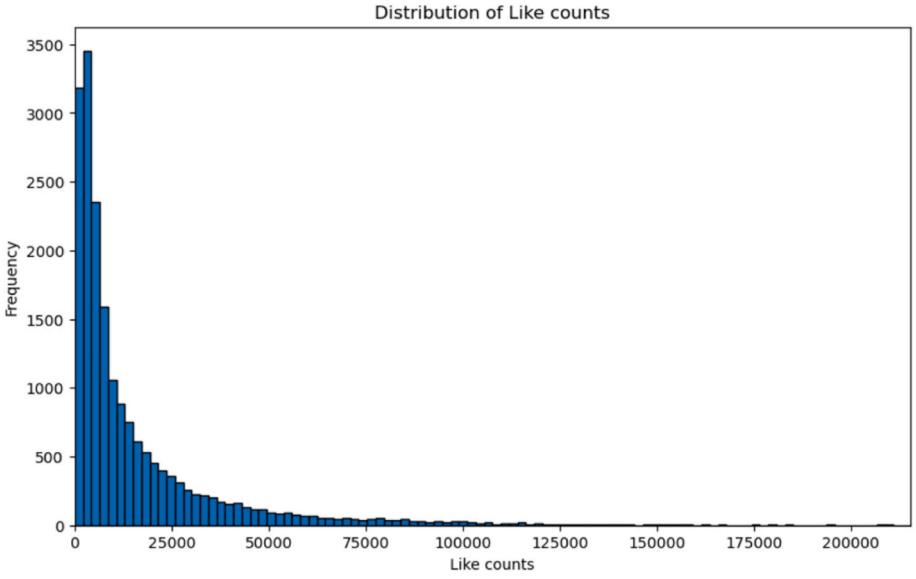


Fig. 3. Like count distribution of collected instagram sponsored posts.

The model was trained using the Adam optimizer and a categorical cross-entropy loss function. Five-fold cross-validation was applied for model training, with a batch size of 32 over 20 epochs. Considering model efficiency, the size of the dense layer was restricted to within 100 dimensions, and finally set to 12 dimensions, which yielded better performance, achieving an average accuracy of 79.43 % on the training set, 77.04 % on the validation set, and 76.88 % on the test set. The result demonstrated the satisfactory performance of using the image visual features of sponsored posts for popularity prediction. The 12-dimensional dense layer were subsequently extracted as visual features of the images.

3.3.2. *Image topic feature extraction*

To extract the topic features of the post images, this study utilized Google Vertex AI Vision APIs for image-to-text conversion to convert each post image into a textual caption in English. An example is shown in Fig. 4.

Subsequently, the LDA (Blei et al., 2003) was employed to uncover the underlying topics of the image captions and estimate the probability of topics within each caption. LDA is widely used in social content analysis and online review analysis to explore semantic content and uncover latent topics (Jelodar et al., 2019; Mahdikhani, 2022). LDA operates on the premise that documents are composed of a mixture of latent topics, with each topic represented by a unique distribution of

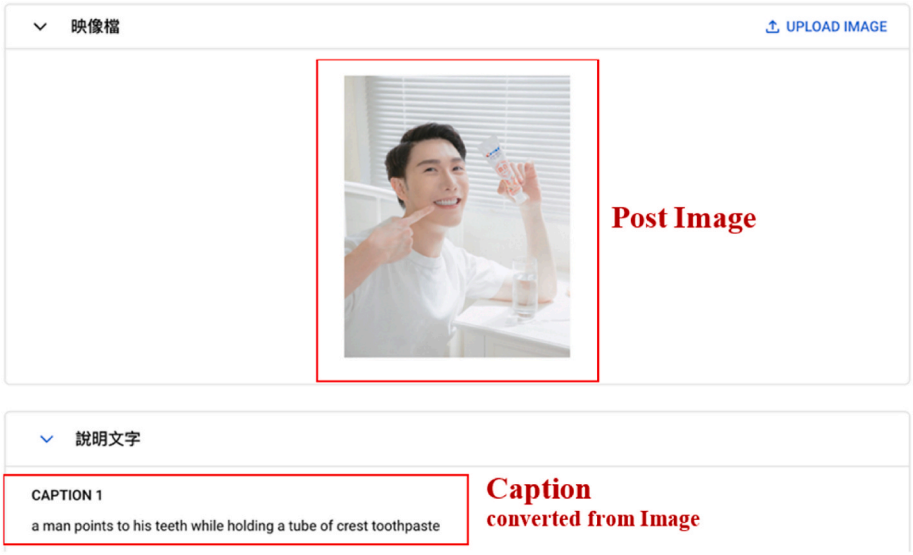


Fig. 4. Google Vertex AI vision APIs for image-to-text conversion.

terms. By analyzing a large corpus, LDA identifies the probabilistic relationships between terms and topics, as well as between topics and documents. Topics can be interpreted and labeled based on the terms they encompass. A document can be semantically characterized by the probability of topics. To determine the optimal number of topics, the coherence score was used as a metric. Coherence measures the semantic similarity of terms within topics, with higher scores indicating stronger term associations and clearer topic concepts, typically better model performance (Röder et al., 2015).

This study utilized the LdaModel package from the Gensim library in Python to extract topics from textual captions converted from images. After text preprocessing, the captions were transformed into a caption-term frequency matrix as input data. The parameter passes was set to 10, alpha and eta were set to auto, chunksize was set to 100, and iterations was set to 630. The relationship between the number of topics and the coherence score is shown in Fig. 5, with the optimal number of topics determined to be 11 based on a higher coherence score.

The 11 topics, along with their constituent terms and their probabilities, and names derived accordingly, are shown in Table 2. These topics are ranked in descending order of intensity. Taking the strongest topic, Topic 1, as an example, it consists of key terms such as standing, picture, group, people, child, and posing, primarily describing a group of people (possibly including children) standing or posing in a certain scene, with possible background elements such as trees or a couch. This led to its naming as "Group Posing Scene." Topic 2, including key terms

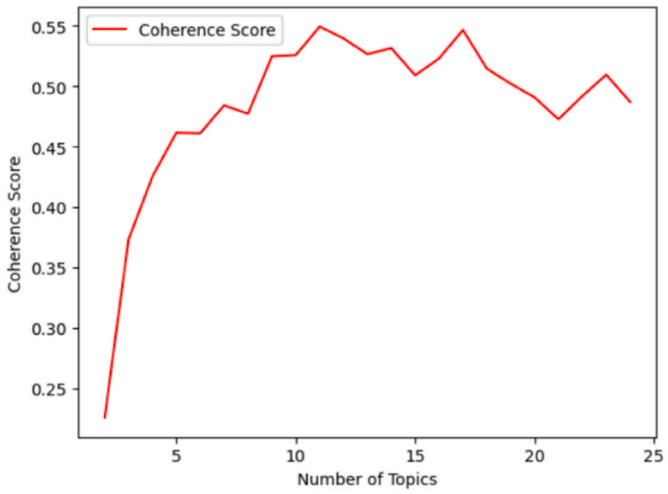


Fig. 5. Coherence score vs. number of topics.

Table 2
Topics of sponsored post images.

Topic	Terms	Name
1	standing(0.133), picture(0.098), group (0.094), people(0.094), child(0.087), posing (0.064), couch(0.036), background(0.022), tree(0.016), dancing(0.014)	Group Posing Scene
2	table(0.236), plate(0.178), food(0.157), eating (0.088), including(0.024), bear(0.019), hello (0.011), kitty(0.009), old(0.009), advertisement(0.008)	Dining Scene with Decorative Elements
3	sits(0.187), glass(0.125), bottle(0.102), wooden(0.054), cake(0.039), dessert(0.034), filled(0.034), drink(0.033), container(0.026), cola(0.023)	Casual Dining with Drinks and Desserts
4	bowl(0.208), food(0.044), close(0.044), large (0.042), egg(0.042), topped(0.030), fruit (0.027), pan(0.025), taipei(0.025), full(0.018)	Bowl Dishes with Toppings
5	face(0.163), yellow(0.074), word(0.069), laying(0.056), making(0.052), phone(0.047), flower(0.041), strawberry(0.038), funny (0.037), window(0.021)	Playful Portraits and Objects
6	holding(0.231), person(0.141), food(0.061), hand(0.047), cup(0.035), box(0.033), cream (0.031), ice(0.031), bag(0.028), family(0.028)	Persons Holding Food or Objects
7	girl(0.167), little(0.118), boy(0.108), pink (0.052), blue(0.043), red(0.037), dress(0.035), young(0.034), sweater(0.023), pant(0.023)	Children in Colorful Outfits
8	sitting(0.172), mask(0.081), microphone (0.064), floor(0.045), giving(0.029), singing (0.019), carrying(0.019), chair(0.018), cartoon(0.017), behind(0.016)	Indoor Performance or Presentation
9	noodle(0.080), cat(0.058), machine(0.049), heart(0.041), pouring(0.040), air(0.033), soup (0.031), airplane(0.027), rock(0.023), house (0.023)	Flowing Flavors and Travel Delights
10	woman(0.192), man(0.149), wearing(0.115), say(0.101), shirt(0.057), sign(0.042), stand (0.035), white(0.034), black(0.027), building (0.017)	Individuals and Interactions
11	chinese(0.281), writing(0.263), baby(0.065), green(0.038), hat(0.037), jacket(0.028), bed (0.025), wall(0.019), bunch(0.018), speaking (0.015)	Chinese Subtitles

like table, plate, food, and eating, strongly suggests a dining or food-related context. Considering the other less dominant terms like bear, hello, and kitty, which might indicate a playful or decorative element, Topic 2 was named "Dining Scene with Decorative Elements."

Notably, it can be observed that many of the top-ranked topics are related to food, indicating that the presentation of food is highly prevalent in the visual content of sponsored posts. This aligns with the phenomenon where younger generations tend to use social media platforms rather than traditional search engines to discover food and restaurants. Consequently, food-related content generates high engagement, and the rise of food influencers and the sponsored post market has followed suit (TechOrange & KOL Radar, 2023).

Next, the constructed LDA model is used to estimate the probability of each image caption belonging to various topics. Table 3 presents some examples. Through this method, the images were converted into textual captions, which were then transformed into probability distributions over topics. These probability values on topics serve as the final extracted topic features of images, representing their semantic content. This allows for a more detailed analysis of successful sponsored post images, thereby providing a clear understanding of the visual elements preferred by the public. This offers valuable insights for brands in their creative design and content strategy.

3.4. Text topic feature extraction

The textual content of posts is often rich and diverse. To better analyze the impact of textual content on popularity, this study employed BERTopic (Grootendorst, 2022) for modeling the textual content of each post as its relevance to various topics, serving as the textual topic features of the posts. Considering that textual content may include multiple languages, using a multilingual-compatible topic model is particularly important. BERTopic demonstrates strong advantages and flexibility in handling multilingual data.

BERTopic is a topic modeling technique built on the BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2018). It leverages pre-trained BERT models to transform textual content into high-dimensional embedding vector representations that capture semantics. To reduce dimensionality and retain critical features, UMAP (Uniform Manifold Approximation and Projection) is employed, projecting these high-dimensional vectors into a lower-dimensional space suitable for clustering. HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based clustering algorithm, is then used to group similar vectors, forming clusters that represent potential topics. For semantically associating each cluster with a interpretable topic, BERTopic uses CountVectorizer to compute the frequency of words within each cluster, excluding common stop

Table 3
The converted captions and topic probabilities of sponsored post Images.

Image	Converted Caption and Topic Probability
	A woman stands in front of a table full of snacks and a sign that says summer day. Topic 1: 0.0355, Topic 2: 0.1056, Topic 3: 0.0172, Topic 4: 0.1019, Topic 5: 0.0355, Topic 6: 0.1056 Topic 7: 0.0262, Topic 8: 0.0271, Topic 9: 0.0000, Topic 10: 0.4984, Topic 11: 0.0295
	A group of people posing for a picture with a teddy bear wearing a t-shirt that says abercrombie & fitch. Topic 1: 0.4094, Topic 2: 0.1493, Topic 3: 0.0141, Topic 4: 0.0160, Topic 5: 0.0133, Topic 6: 0.0437 Topic 7: 0.0215, Topic 8: 0.0222, Topic 9: 0.0000, Topic 10: 0.2783, Topic 11: 0.0242
	A man in a green shirt holds up several bottles with Chinese writing on them. Topic 1: 0.0358, Topic 2: 0.0249, Topic 3: 0.0968, Topic 4: 0.0197, Topic 5: 0.0164, Topic 6: 0.1333 Topic 7: 0.0265, Topic 8: 0.0274, Topic 9: 0.0000, Topic 10: 0.3411, Topic 11: 0.2682
	A bunch of strawberries and sandwiches on a table with boxes of strawberries in the background. Topic 1: 0.1253, Topic 2: 0.1134, Topic 3: 0.0188, Topic 4: 0.0214, Topic 5: 0.1905, Topic 6: 0.2311 Topic 7: 0.0288, Topic 8: 0.0297, Topic 9: 0.0108, Topic 10: 0.1115, Topic 11: 0.1187

Table 4
Topics of sponsored post textual content.

Topic	Representation Terms	Name
1	吃(0.02), 好吃(0.01), 麵(0.01), 茶(0.01), 口味(0.01), 草莓(0.01), 時間(0.01), 搭配(0.01), 口感(0.01)	美食推薦 Food Recommendations
2	姐(0.02), 阿母(0.02), 一起(0.01), 一個(0.01), 今天(0.01), 玩(0.01), 喜歡(0.01), 巨人(0.01)	親朋好友遊樂 Fun Activities with Family and Friends
3	髮(0.06), 植村秀(0.02), 穿(0.02), 敏感(0.02), 洗(0.02), 舒酸定(0.02), 肌膚(0.01), 妝(0.01), 牙刷(0.01), 牙齒(0.01)	個人護理與美妝 Personal Care and Beauty
4	sony(0.05), 相機(0.05), 手機(0.04), 拍(0.04), zv(0.04), 1f(0.03), 使用(0.03), 操作(0.03), 方便(0.03)	拍攝器材 Photography Equipment
5	podcast(0.13), 完整版(0.12), yt(0.12), 少年(0.09), 人生(0.08), 心理(0.06), booktender(0.06), 課程(0.05), 命理(0.05), 諮商(0.04)	個人成長與學習 Personal Growth and Learning
6	敏感(0.04), always(0.03), 衛生棉(0.03), 肌膚(0.03), 月經(0.03), 乾癢(0.03), 結婚(0.02), 老婆(0.02), 巧克力(0.02)	經期護理與情感支持 Menstrual Care and Emotional Support
7	eat(0.14), want(0.14), dying(0.14), chicken(0.11), find(0.09), tbsp(0.09), interviewing(0.08), 1tbs(0.07), tsp(0.07), rice(0.06)	美食探索與烹飪食譜 Food Exploration and Cooking
8	冰(0.10), 癒(0.09), 蛋(0.08), 藍(0.08), 砂糖(0.07), 雞蛋餅(0.07), 蛋白(0.07), 三角形(0.06), 天氣(0.06), 冰球(0.06)	甜點飲品 Desserts and Beverages
9	聖誕節(0.07), 聖誕(0.06), 咖啡(0.06), 星耀(0.05), 巴黎(0.04), 榛果(0.04), 哈哈哈(0.04), 今年(0.04), 烤(0.03), 甜點(0.03)	節慶活動 Festival Activities
10	政治(0.26), 選舉(0.17), 節目(0.12), 確啦(0.12), 失和(0.12), 女方(0.12), 共識(0.12), 男方(0.12), 政見(0.11), 不正(0.11)	政治議題 Political Issues
11	西亞(0.34), live(0.21), 馬(0.20), podcast(0.18), 百靈果(0.15), menara(0.10), 首站(0.10), accupass(0.10), 接機(0.10), 場地(0.09)	巡迴展演與接機活動 Tour Performances and Airport Welcoming
12	want(0.51), 她贏(0.46), win(0.44), 希望(0.26), maybe(0.25), saying(0.25), 錯話(0.25), thinking(0.25), assertive(0.25), teach(0.25)	人際溝通與反思 Interpersonal Communications and Reflection
13	火車(0.33), adventures(0.23), episode(0.23), train(0.22), 奇遇(0.22), 高鐵(0.21), 列車(0.20), 自強(0.19), 一邊(0.17), 電影(0.16)	冒險旅行 Adventure Travel
14	Black pink official(1.13), black pink(1.10), pink venom(0.50), 演唱(0.43), 昨天(0.33), typa(0.27), shut(0.25), 終回(0.25), 兩段(0.24)	韓國流行女團 K-pop Girl Groups

words, and applies class-based Term Frequency-Inverse Document Frequency (c-TF-IDF) to assign importance scores to words within a specific cluster. BERTopic does not require a predefined number of topics. Instead, it dynamically determines the number of clusters based on the min_cluster_size parameter of HDBSCAN, allowing flexibility in topic discovery.

This study utilized the paraphrase-multilingual-MiniLM-L12-v2 model supported by the Python BERTopic package for multilingual topic analysis. The tokenizer, jieba, for Chinese text segmentation, was used in the CountVectorizer. The coherence score was used to evaluate the effectiveness of different cluster (topic) numbers generated under various min_cluster_size settings of 5, 10, 15, 20, 30, and 50. Finally, 14 topics were produced when the min_cluster_size was set to 5 or 10, achieving the highest coherence score. The representation terms, their c-TF-IDF values, and names derived accordingly are shown in Table 4. Finally, for each post text, the probabilities of 14 topics were inferred, resulting in a 14-dimensional textual topic feature vector.

3.5. Popularity prediction model establishment

This study utilized the four sets of features extracted from sponsored posts—namely, 24-dimensional influencer features, 12-dimensional image visual features, 11-dimensional image topic features, and 14-dimensional textual topic features—both individually and in combination as predictive variables. Various methods, including statistical methods, machine learning, and neural network models, were employed to predict the popularity of sponsored posts. The models used include logistic regression (LR), random forests (RF), support vector machines (SVM), XGBoost (XGB), deep neural networks (DNN), and a hard voting-based ensemble model that integrates the aforementioned models (EM). For model construction, Python libraries and packages were used. For analysis, the features were normalized in order to eliminate scale differences. The five-fold cross-validation was applied to generate training and testing results. Additionally, an 80-20 random split of the training data was used to tune the parameters. This approach ensures the reliability of the analysis results. The metrics for evaluating the prediction performance were:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3.1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.3}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.4}$$

where TP, FP, TN, and FN means the number of true positive, false positive, true negative, and false negative examples, respectively.

3.6. Feature effect analysis

To clarify the impact of various features on the popularity of sponsored posts and generate business insights, this phase involves two levels of analysis. The first level examined the effects of the four extracted feature types on the popularity of sponsored posts. To achieve this, the study adopted Taguchi experiments. The second level delved deeper into analyzing how individual features within each feature type influence the popularity of sponsored posts. For this purpose, the study interpreted the impact of individual features using the odds ratios derived from logistic regression analysis.

3.6.1. Effects of feature types

To further understand the effect of four different feature types on the popularity of sponsored posts, this study employed the Taguchi experiments (Taguchi et al., 2005). Developed by Genichi Taguchi, the Taguchi experiments optimize quality systems by using orthogonal arrays and the signal-to-noise (S/N) ratio to evaluate the contribution of system factors to system quality and determine the optimal factor settings for system quality optimization. Orthogonal arrays enable utilizing an experiment design of a minimum number of factor combinations to evaluate the main effect of individual factors with minimal effect confounding. Also, orthogonal arrays design the factorial experiments to be conducted under various noise conditions in order to ensure the analysis results are robust and insensitive to noise. The S/N ratio is used to evaluate system performance under different experiments of factor combinations, measuring both the quality signal and the variability introduced by noise. The higher the S/N ratio, the better the system quality. The critical factors affecting system quality can be identified by evaluating and comparing the S/N ratios of different experiments.

This study defined four system factors that may affect post popularity: image visual feature set (Factor A), image topic feature set (Factor B), textual topic feature set (Factor C), and influencer feature set (Factor D). The L8 orthogonal array was employed in this analysis, as shown in Fig. 6. The L8 orthogonal array has seven orthogonal columns for accommodating up to seven two-level factors and creates eight

Experiment	Factor							Noise						S/N
								LR	RF	SVM	XGB	DNN	EM	
	A	B	C	D	-	-	-	Observations						
1	1	1	1	1	1	1	1	y_{11}	y_{12}	y_{13}	y_{14}	y_{15}	y_{16}	S/N_1
2	1	1	1	2	2	2	2	y_{21}	y_{22}	y_{23}	y_{24}	y_{25}	y_{26}	S/N_2
3	1	2	2	1	1	2	2	y_{31}	y_{32}	y_{33}	y_{34}	y_{35}	y_{36}	S/N_3
4	1	2	2	2	2	1	1	y_{41}	y_{42}	y_{43}	y_{44}	y_{45}	y_{46}	S/N_4
5	2	1	2	1	2	1	2	y_{51}	y_{52}	y_{53}	y_{54}	y_{55}	y_{56}	S/N_5
6	2	1	2	2	1	2	1	y_{61}	y_{62}	y_{63}	y_{64}	y_{65}	y_{66}	S/N_6
7	2	2	1	1	2	2	1	y_{71}	y_{72}	y_{73}	y_{74}	y_{75}	y_{76}	S/N_7
8	2	2	1	2	1	1	2	y_{81}	y_{82}	y_{83}	y_{84}	y_{85}	y_{86}	S/N_8

Fig. 6. Taguchi experiment design.

experiments of different factor combinations. The four system factors were set as two levels, representing whether the feature set is included (Level 1) or excluded (Level 2) in the construction of the prediction model, and were arranged into the first four columns. The six prediction methods (i.e., LR, RF, SVM, XGB, DNN, and EM) that can affect the prediction performance were treated as noise. The predictions of each experiment have to be conducted using all six methods. Consequently, each experiment has six observations, each was the average F1 score obtained from five-fold cross-validation under the factor combination and the prediction method. Since a higher F1 score indicates better performance, the "larger-the-better" S/N ratio was adopted for the evaluation of each experiment, as expressed in Equation (3.5).

$$S/N_i = -10 \bullet \log\left(\frac{1}{n_i} \bullet \sum_{j=1}^{n_i} \frac{1}{y_{ij}^2}\right) \tag{3.5}$$

where n_i is the number of observations of experiment i; y_{ij} is the value of jth observation of experiment i.

Using the S/N ratio from each experiment, the effect gain (ρ) of each factor can be calculated (Equation (3.6)). A positive effect gain means that the factor positively contributes to the popularity prediction of sponsored posts, and the higher the more significant. This helps explain the importance of the four different feature types in the popularity of sponsored posts.

$$\rho = \overline{S/N}^1 - \overline{S/N}^2 \tag{3.6}$$

where $\overline{S/N}^1$ is the average S/N ratio over the experiments including the factor (Level 1); $\overline{S/N}^2$ is that excluding the factor (Level 2).

3.6.2. Effect of individual features

In the Taguchi experiments, the S/N ratio identified which feature types contribute most to the popularity of sponsored posts. To derive actionable business insights, a more detailed understanding of each individual feature's impact and magnitude is required.

Since the set of image visual features extracted by ResNet50 holistically represents the overall visual content of an image, it is challenging and meaningless to interpret its individual features. Therefore, the individual image visual features were not further discussed. For the features in the remaining three types, the odds ratio from logistic regression analysis was further used to interpret the direction of impact on post popularity. The odds ratio, a statistical metric, reflects how a unit increase in a feature affects the odds of the target event occurring. Values significantly greater than 1 indicate that an increase in the feature substantially raises the likelihood of the target event occurring, while values less than 1 suggest a negative impact. This provides a clear and intuitive explanation of individual features' influence on the target event.

Table 5
Popularity prediction performance.

Predictor	Model	Accuracy	Precision	Recall	F1-Score
Influencer Features	RF	83.18 %	89.99 %	74.67 %	81.60 %
	XGB	83.24 %	90.23 %	74.58 %	81.64 %
	LR	82.03 %	87.20 %	75.10 %	80.68 %
	SVM	83.18 %	90.08 %	74.58 %	81.59 %
	DNN	83.37 %	90.39 %	77.24 %	81.79 %
	EM	83.52 %	90.25 %	75.18 %	82.01 %
Image Visual Features	RF	81.43 %	83.03 %	79.03 %	80.97 %
	XGB	79.99 %	81.01 %	78.39 %	79.67 %
	LR	81.88 %	84.46 %	78.14 %	81.16 %
	SVM	82.07 %	85.13 %	77.75 %	81.26 %
	DNN	81.63 %	83.97 %	78.25 %	80.34 %
	EM	82.02 %	84.64 %	78.29 %	81.31 %
Image Topic Features	RF	69.62 %	68.83 %	71.77 %	70.25 %
	XGB	69.62 %	69.51 %	70.06 %	69.75 %
	LR	68.51 %	68.13 %	69.72 %	68.89 %
	SVM	68.74 %	67.21 %	73.30 %	70.11 %
	DNN	68.53 %	66.38 %	75.10 %	70.44 %
	EM	69.68 %	68.15 %	73.94 %	70.91 %
Text Topic Features	RF	64.90 %	67.44 %	57.65 %	62.13 %
	XGB	65.39 %	67.74 %	58.81 %	62.92 %
	LR	66.18 %	67.16 %	63.62 %	65.29 %
	SVM	67.10 %	68.72 %	62.90 %	65.64 %
	DNN	67.42 %	69.80 %	62.22 %	65.77 %
	EM	67.36 %	69.20 %	62.60 %	65.71 %
All Features	RF	86.14 %	90.45 %	80.81 %	85.35 %
	XGB	85.42 %	87.37 %	82.81 %	85.02 %
	LR	85.22 %	87.88 %	81.71 %	84.67 %
	SVM	86.31 %	91.40 %	80.17 %	85.40 %
	DNN	86.72 %	89.69 %	82.99 %	86.20 %
	EM	86.29 %	89.22 %	82.56 %	86.03 %

4. Results

4.1. Popularity prediction

To predict the popularity of sponsored posts (high or low), five feature configurations were utilized: four individual feature sets and a combined feature set. For model construction, Python libraries and packages were used. The default values were applied for the parameter settings of RF, XGB, and LR. For SVM, the Radial Basis Function kernel was used, and Grid Search was employed to determine the optimal settings for the gamma and cost parameters.

DNN was explored with various architectures, and the best-performing one was selected. In the prediction model based on the 24-dimensional influencer features, a 1D-CNN model and deeper architectures incorporating multiple fully connected layers were experimented.

However, these more complex and deeper models did not yield significant improvements in prediction performance compared to single-hidden-layer models. Consequently, the final model architecture was simplified to include a single hidden layer with 10 nodes, employing the ReLU activation function. The output layer consists of a single node with a sigmoid activation function. During training, the Adam optimizer with a learning rate of 0.001 and binary cross-entropy were selected to guide optimization. Models utilizing 12-dimensional image visual features, 11-dimensional image topic features, 14-dimensional text topic features, and a combination of all features (a total of 61 dimensions) were developed using a 1D-CNN architecture. The architecture comprises two convolutional layers, each equipped with 64 kernels of size 3 and activated using the ReLU function. To mitigate overfitting, a dropout layer with a rate of 0.5 was added after the convolutional layers. Subsequently, a max-pooling layer with a pool size of 2 was applied for feature extraction, followed by a flatten layer to form a vector. The vector was then passed through a fully connected layer with 100 nodes, activated by the ReLU function, and a dropout layer with the rate of 0.5. Finally, a single-node output layer with a sigmoid activation function was used for binary classification. Model training was conducted using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy as the loss function.

Table 5 presents the classification performance across different feature sets and models, with F1-score as the primary evaluation metric. Overall, models combining all features achieved the highest F1-Scores, with DNN (86.20 %) and SVM (85.40 %) delivering the best results, highlighting the value of feature integration in improving the balance between precision and recall.

Among individual feature sets, image visual features stand out as a particularly important predictor. Although their F1-scores (ranging from 79.67 % to 81.31 %) are slightly lower than those of influencer features, their role in modeling is noteworthy. Visual features serve as direct stimuli that resonate with audiences, even if their influence on the prediction of post popularity may not be explicitly explainable. For instance, aesthetic qualities, color schemes, or visual appeal can significantly impact audience engagement, making these features valuable for predicting the success of sponsored posts on Instagram. EM achieved the highest F1-score (81.31 %) for this feature set, closely followed by SVM (81.26 %), reflecting the consistent contribution of visual elements to model performance. Image topic features exhibited even lower F1-scores, between 68.89 % and 70.91 %. The EM model performed best on this feature set, with an F1-score of 70.91 %, indicating moderate discriminative power.

In contrast, text topic features demonstrated the lowest F1-scores across all feature sets, ranging from 62.13 % to 65.77 %. This result indicates that text content is not a critical determinant of post success on Instagram. The abstract and often inconsistent nature of textual information in posts may dilute its predictive power. DNN achieved the highest F1-score for this feature set (65.77 %), slightly outperforming EM (65.71 %). These findings suggest that while text provides context, it is far less influential than visual features or influencer-related attributes in determining the popularity of sponsored posts.

In summary, the analysis confirmed that combining these diverse feature types yielded the highest discriminatory capability, achieving an accuracy of over 85 % in predicting whether a sponsored post would gain popularity. Image visual features are highly valuable despite their abstract nature, as they capture the visual appeal that drives engagement, an essential factor for predicting post success on Instagram. Meanwhile, the weak performance of text topic features underscores that textual content plays a secondary role in influencing engagement. This highlights the importance of prioritizing visual elements and influencer characteristics in optimizing strategies for sponsored Instagram posts.

4.2. Effects of feature types

To further explore the impact of the four feature types on sponsored post popularity, this study employed the Taguchi experiments. This approach can provide insights into the relative importance of these features to post popularity. The experimental data and the S/N ratio calculations are presented in Table 6, while the effect gains of the four types of features are illustrated in Fig. 7. It is revealed that image visual features (A), image topic features (B), and influencer features (D) are of positive effect gains, which means a positive contribution to improving prediction accuracy, whereas text topic features (C) exhibit the weakest or even negative effects on model performance. Among these, the significance of image visual features and influencer features is particularly pronounced, underscoring the critical role of post images and the personal characteristics of influencers in driving the popularity of sponsored posts.

The findings highlight the importance of focusing on image visual features and influencer features when analyzing successful sponsored posts. However, due to the interpretive challenges associated with visual features, this study utilized image topic features to provide additional clarification. Although the impact of image topic features is relatively smaller, they offer valuable insights by supplementing the limitations of visual features and providing more concrete semantic meanings to the images, which aids in interpreting the model results. Additionally, influencer features, including fields of expertise, gender, and cross-platform performance metrics, significantly influence the success of sponsored posts. These findings suggest that brands should carefully evaluate influencer attributes when selecting collaboration partners to ensure their style and target audience align closely with the brand's image and market strategy.

4.3. Effect of individual features

This study further employed an odds ratio analysis using logistic regression to explore in greater detail the mechanisms through which the individual features impact the popularity of sponsored posts. The

Table 6
Observations and S/N ratios of Taguchi experiments.

					Noise						
					LR	RF	SVM	XGB	DNN	EM	
Experiment	Factor				Observations						S/N
	A	B	C	D							
1	1	1	1	1	84.67 %	85.35 %	85.40 %	85.02 %	86.20 %	86.03 %	38.63
2	1	1	1	2	81.02 %	84.45 %	81.64 %	84.18 %	82.27 %	82.36 %	38.34
3	1	2	2	1	84.33 %	85.25 %	84.59 %	84.36 %	85.86 %	85.80 %	38.60
4	1	2	2	2	81.16 %	80.97 %	81.26 %	79.67 %	80.34 %	81.31 %	38.15
5	2	1	2	1	80.62 %	82.03 %	81.63 %	80.98 %	82.27 %	83.19 %	38.25
6	2	1	2	2	68.89 %	70.25 %	70.11 %	69.75 %	70.44 %	70.91 %	36.91
7	2	2	1	1	80.55 %	79.71 %	81.24 %	80.45 %	80.78 %	81.48 %	38.14
8	2	2	1	2	65.29 %	62.13 %	65.64 %	62.92 %	65.77 %	65.71 %	36.19

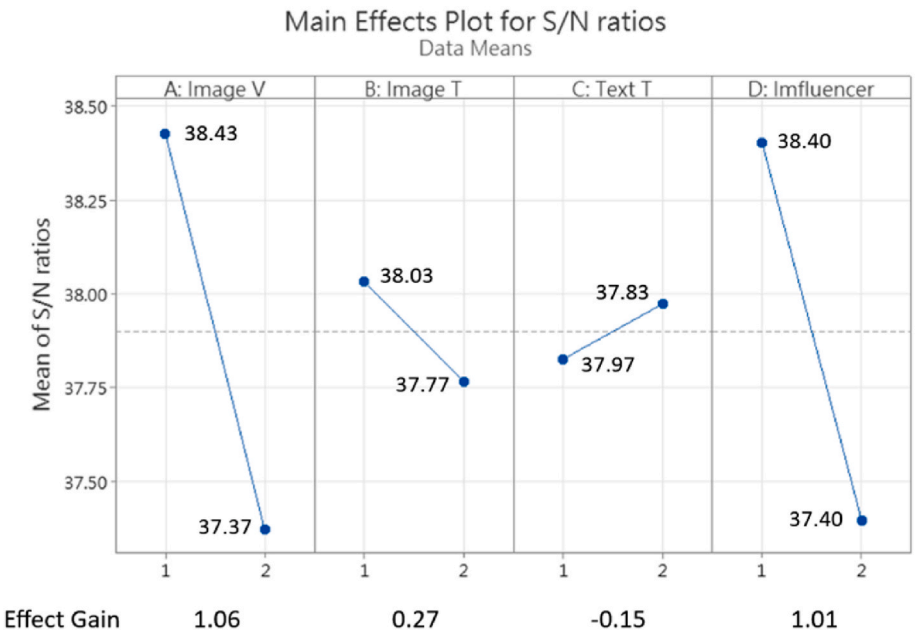


Fig. 7. Main effect plot and effect gain of factors.

Table 7
Odds ratios of individual features.

Influencer Features				Image Topic Features		Text Topic Features	
Feature	Odds Ratio	Feature	Odds Ratio	Feature	Odds Ratio	Feature	Odds Ratio
• IG Post Counts	0.4430	• Lifestyle	1.6183	• Group Posing Scene	5.1374	• Food Recommendations	0.3109
• IG Following Counts	1.1440	• Fashion	1.3995	• Dining Scene with Decorative Elements	0.0493	• Fun Activities with Family and Friends	1.9026
• IG Follower Counts	9.2223	• Sports	0.9604	• Casual Dining with Drinks and Desserts	0.3417	• Personal Care and Beauty	1.6779
• IG Engagement Rates	2.7502	• Travel	1.0889	• Bowl Dishes with Toppings	0.1973	• Photography Equipment	0.9126
• IG View Counts	0.3968	• Affection	0.8939	• Playful Portraits and Objects	1.3382	• Personal Growth and Learning	4.4975
• FB Follower Counts	1.0629	• Technology	0.7870	• Persons Holding Food or Objects	0.1400	• Menstrual Care and Emotional Support	4.9043
• FB Engagement Rates	0.8101	• Education	0.2198	• Children in Colorful Outfits	23.6104	• Food Exploration and Cooking	0.8243
• FB View Counts	1.1000	• Entertainment	0.7897	• Indoor Performance or Presentation	1.3198	• Desserts and Beverages	0.2752
• YT Subscriber Counts	1.0098	• Beauty	0.4572	• Flowing Flavors and Travel Delights	0.2313	• Festival Activities	0.8074
• YT Engagement Rates	0.4497	• Food	0.3739	• Individuals and Interactions	12.3740	• Political Issues	0.4281
• YT View Counts	0.7326	• Male	0.8805	• Chinese Subtitles	3.3490	• Tour Performances and Airport Welcoming	0.2470
		• Female	1.6217			• Interpersonal Communications and Reflection	0.3594
		• Non-Gendered	0.6965			• Adventure Travel	0.5016
						• K-pop Girl Groups	2.0719

findings can help brands optimize the design and content strategies of their sponsored posts to enhance audience appeal and engagement, thereby improving the overall market effectiveness of sponsored content. The odds ratios of individual features are shown in Table 7, and explained in the following subsections.

4.3.1. Analysis of influencer features

The analysis of influencer features revealed that the Instagram Follower Counts (Odds Ratio: 9.2223) and Instagram Engagement Rates (Odds Ratio: 2.7502) are the primary positive factors influencing the popularity of sponsored posts. A larger follower base on Instagram suggests that an influencer has a more loyal and extensive audience, while a higher engagement rate indicates active interaction with users, increasing post visibility and ultimately leading to a higher number of likes and greater post popularity. These findings suggest that brands should prioritize collaborations with influencers who have a substantial follower base and high engagement rates on Instagram. Additionally, other metrics, such as the Facebook Follower Counts (Odds Ratio: 1.0629), Facebook View Counts (Odds Ratio: 1.1000), and YouTube Subscriber Counts (Odds Ratio: 1.0098), also exhibit a positive impact. This indicates that an influencer's activity and presence on multiple social media platforms can serve as valuable indicators of their overall influence. Given the distinct characteristics and audience demographics of different platforms, influencers who actively manage multiple

platforms provide brands with opportunities to maximize exposure. Consequently, brands may consider cross-platform content distribution for sponsored posts to enhance visibility and reach a broader potential consumer base.

Moreover, influencers specializing in Lifestyle (Odds Ratio: 1.6183), Fashion (Odds Ratio: 1.3995), and Travel (Odds Ratio: 1.0889) are more likely to attract attention. This may reflect Instagram users' preference for these fields of influencers and content, as their posts tend to be more relatable and engaging for a general audience. In contrast, influencers specializing in Education exhibit a lower odds ratio (Odds Ratio: 0.2198), suggesting that users are less inclined to engage with educational content on social media, favoring more lighthearted and lifestyle-oriented content instead. Additionally, influencers in Beauty (Odds Ratio: 0.4572) and Food (Odds Ratio: 0.3739) show odds ratios below 1, indicating a lower likelihood of their sponsored posts achieving high popularity. This could be due to the high number of influencers in these categories, providing audiences with more options and making it harder for individual influencers to stand out. Alternatively, the similarity of content within these fields may necessitate greater innovation or brand differentiation to capture user attention. Notably, Female influencers (Odds Ratio: 1.6217) are more likely to produce content that enhances post popularity, suggesting that Instagram audiences tend to prefer content created by female influencers. On the other hand, influencers with excessive Instagram Post Counts or a sole focus on maximizing Views (Odds Ratios: 0.4430 and 0.3968, respectively) do not significantly enhance post popularity, highlighting that content quality is more crucial than quantity.

4.3.2. Analysis of image topic features

In terms of image topic features, Children in Colorful Outfits (Odds Ratio: 23.6104) and Individuals and Interactions (Odds Ratio: 12.3740) are the most attractive, indicating that authentic emotional interactions and visually appealing, vibrant imagery generate positive emotions and a sense of warmth, thereby increasing popularity. Additionally, Group Posing Scene (Odds Ratio: 5.1374) contributes to audience engagement by portraying social interactions, which may evoke a sense of connection among viewers and enhance their identification with the image content. Furthermore, the presence of Chinese Subtitles (Odds Ratio: 3.3490) enhances readability, particularly in the context of the Taiwanese market. It is hypothesized that Chinese text within images serves as a focal point, improving content legibility and subsequently attracting user attention.

Conversely, Dining Scene with Decorative Elements (Odds Ratio: 0.0493) and Persons Holding Food or Objects (Odds Ratio: 0.1400) may negatively impact popularity, possibly due to a lack of emotional resonance or an overt marketing intention that deters audience engagement. Similarly, Flowing Flavors and Travel Delights (Odds Ratio: 0.2313), Bowl Dishes with Toppings (Odds Ratio: 0.1973), and Casual Dining with Drinks and Desserts (Odds Ratio: 0.3417) perform poorly, likely because they appear too commonplace or fail to capture users' attention. Notably, the image topics with negative effects are predominantly food-related, suggesting that the prevalence of such images on social media may lead to aesthetic fatigue, reducing their effectiveness in attracting interest. Moreover, close-up shots of people holding objects tend to make commercial intent immediately apparent, potentially diminishing user interest or even triggering aversion.

Overall, the analysis of image topic features highlights successful features of attractiveness, emotional engagement, or strong visual impact, while overly static or uncreative designs should be avoided. Product placement should be integrated naturally into real-life scenarios, creating an authentic usage context that aligns with brand identity while avoiding overt promotional strategies.

4.3.3. Analysis of text topic features

In terms of textual content, the topic of Menstrual Care and Emotional Support (Odds Ratio: 4.9043) exhibits the highest positive impact. This is primarily because its content effectively integrates practical information with emotional support, conveying a sense of understanding and care for women. Such content enhances the authenticity and engagement of posts, enabling female users to perceive the brand as thoughtful and considerate. Additionally, these posts contribute to increasing men's awareness and empathy toward menstrual issues, fostering a more open and inclusive discussion environment. Personal Growth and Learning (Odds Ratio: 4.4975) encompasses topics such as goal setting, learning methods, and mindset development, addressing the contemporary users' desire for self-improvement. Instagram users, particularly from the younger generation, show a strong preference for such inspirational content, as they seek practical knowledge and life insights from social media.

The positive odds ratios of topics K-pop Girl Groups (Odds Ratio: 2.0719) and Fun Activities with Family and Friends (Odds Ratio: 1.9026) reflect users' preferences for entertainment-oriented and lifestyle-related content. The topic of K-pop Girl Groups has gained popularity due to the influence of pop culture and fan communities, aligning with Instagram users' strong interest in celebrity news and trends, as well as the broader enthusiasm for Hallyu entertainment culture. Meanwhile, the topic of Fun Activities with Family and Friends resonates with users by portraying lighthearted and enjoyable everyday moments, evoking emotional engagement. Users are particularly drawn to content that is relatable, warm, and infused with human connection.

Personal Care and Beauty (Odds Ratio: 1.6779) highlights the contemporary youth's, and particularly Instagram's core user demographic's, strong focus on health and appearance. Posts related to skincare and makeup techniques, often shared by influencers, have gained traction by providing practical recommendations that fulfill users' demand for product information.

Conversely, the topic of Tour Performances and Airport Welcoming (Odds Ratio: 0.2470) primarily attracts a niche fan base, as such posts mainly serve as event records or promotional materials and often lack interactive elements. For the broader Instagram user base, this type of content may appear overly formulaic and fail to evoke engagement. Similarly, the topic of Interpersonal Communications and Reflection (Odds Ratio: 0.3594) typically focuses on influencers' personal experiences and life stories. The relatively serious tone of the writing makes it difficult for the general audience to relate, thereby limiting engagement. The topic of Political Issues (Odds Ratio: 0.4281) tends to be associated with strong ideological positions, attracting a fixed audience but struggling to appeal to a broader user base. The topic of Adventure Travel (Odds Ratio: 0.5016), while inherently attractive, faces challenges in effectively conveying the uniqueness of such experiences through text alone. Many Instagram users may find it difficult to immerse themselves in these narratives, resulting in a sense of detachment from the content.

The food-related topics such as Desserts and Beverages (Odds Ratio: 0.2752) and Food Recommendations (Odds Ratio: 0.3109) primarily consist of descriptions of taste, texture, and ingredients. However, the frequent repetition of similar adjectives may lead to reader fatigue, and the inability to fully convey sensory experiences through text diminishes their impact. The topic of Food Exploration and Cooking (Odds Ratio: 0.8243) demonstrates relatively higher effectiveness, as it incorporates elements such as cooking processes, ingredient combinations, creative recipes, and the exploration of new flavors, adding a layer of novelty and engagement.

Overall, textual content that incorporates emotional care, inspirational learning, and strong entertainment value aligns most effectively with Instagram users' preferences. Content strategies that integrate innovative elements and contextual storytelling can further enhance post popularity and engagement. Conversely, highly niche topics, overly repetitive topics, or influencer-centered content—while appealing to specific audiences—may struggle to generate widespread traction on a mass-market platform.

5. Discussions

This study highlighted the critical role of visual elements and influencer characteristics in the success of sponsored posts on Instagram. Compared to textual content, visual features and influencer attributes have a stronger impact on post popularity. Therefore, brands and influencers should prioritize visual appeal and collaboration choices to maximize content reach and engagement.

Visual elements serve as the first gateway to capturing audience attention. Combined with Image topic analysis, it reveals that high-quality, emotionally resonant images drive more interactions. For instance, vibrant colors and emotionally engaging scenes—such as children in bright clothing or interactive group settings—tend to attract more attention, whereas simple food plating or static product images often underperform due to a lack of emotional connection. Additionally, high-quality visuals showcasing product usage and transformations quickly capture interest, particularly in beauty and fashion-related posts. To enhance authenticity, brands and influencers should avoid overly commercialized product photography and instead incorporate real-life scenarios and genuine interactions. For example, food-related content featuring human interactions or narrative formats can significantly boost engagement.

Follower count and engagement rate are key factors influencing sponsored post performance, highlighting the need to assess an influencer's audience loyalty and activity. Additionally, cross-platform influence is crucial—data from Facebook and YouTube show that influencers with a multi-platform presence achieve broader reach and greater impact. Brands should consider partnering with influencers who manage multiple platforms and leveraging cross-channel distribution to enhance visibility. Influencer performance also varies by content category and demographic factors. Those specializing in lifestyle, fashion, and travel tend to attract more attention, reflecting Instagram users' preferences. Moreover, female influencers generally achieve higher post popularity, likely due to stronger audience engagement and alignment with visually driven content such as fashion and lifestyle. In contrast, influencers in education, food, and beauty face intensified competition due to market saturation. To stand out, they should adopt innovative formats and unique presentation styles to enhance audience appeal.

While visual content is the primary driver of post performance, certain types of textual content can also enhance engagement. Text related to emotional well-being, personal growth, and entertainment tends to attract higher interaction, reflecting Instagram users' preference for storytelling, emotional resonance, and practical value. In contrast, political discussions, highly personal reflections, or text-only food recommendations with little context often receive less attention. Thus, brands and influencers should incorporate emotional appeal and narrative techniques to improve readability and engagement.

As a visually driven platform, Instagram places a strong emphasis on aesthetics. This study confirms that the success of sponsored posts hinges primarily on visual content, followed by influencer characteristics, with textual content serving a complementary role in specific contexts. To enhance content creation, partner selection, and cross-platform strategies, brands and influencers should invest in emotionally compelling visuals and targeted influencer marketing. High-quality, creative imagery can significantly boost engagement, while a lighthearted, everyday storytelling approach fosters deeper audience connection, outperforming static, commercialized product displays. By combining three key elements—visual appeal, lifestyle-driven content, and influencer attributes—brands can create high-quality sponsored posts on Instagram, effectively strengthening brand influence and driving commercial value.

Moreover, as trends evolve, brands and influencers must stay attuned to market shifts. Timely identification of emerging trends and continuous content adaptation are essential. Long-term user engagement and strong emotional connections are also crucial for fostering audience loyalty. By aligning marketing strategies with insights from analytical models, influencers and brands can enhance content appeal, increase engagement, and improve overall marketing effectiveness.

6. Conclusions

This study developed predictive and analytic models to assess the popularity of Instagram influencer-sponsored posts, aiding brands in selecting suitable influencers and optimizing the strategies of sponsored content. A large dataset of sponsored posts was collected, encompassing images, text, and influencer attributes. Four types of features were extracted: image visual features (using ResNet50), image topic features (using Image-to-Text Conversion and LDA analysis), textual topic features (using BERTopic analysis), and influencer features as predictive variables. Notably, Image-to-Text Conversion and LDA-extracted image topics were used to complement the interpretability limitations of ResNet50-extracted image visual features.

The models were constructed using Random Forest, XGBoost, Support Vector Machine, Logistic Regression, Deep Neural Networks, and a voting-based ensemble model. Experimental results indicate that integrating multiple feature types yields the best predictive performance, achieving an F1-score exceeding 85 %, confirming that considering multiple feature types in the sponsored posts on Instagram enables a more accurate prediction of their success.

To further clarify the impact of different feature types, Taguchi experiments were conducted to assess their influence on post success. Results show that image visual features exert the greatest impact, reinforcing that Instagram is a visually driven platform, followed by influencer features, image topics, and textual topics. Additionally, an odds ratio analysis using Logistic Regression was employed to uncover the detailed effects of each feature. The results help brands select influencers for product marketing based on key features and provide guidance for both brands and influencers in optimizing visual presentation, content strategy, and copywriting, ultimately enhancing the return on investment in influencer marketing.

6.1. Contributions and implications

This study provided a more comprehensive understanding of the key success factors in influencer marketing and contributed to the existing literature. The contribution and implications are in several ways.

First, this study focused on influencer marketing on social media platforms, with particular attention to the commercial operations of sponsored posts. Unlike previous studies that have examined the success factors of social media posts without specifically targeting influencers or deliberately addressing sponsored content, this study investigates the presentation and commercial mechanisms of sponsored posts. By analyzing the marketing success elements, this study offered a clearer understanding of the characteristics and effectiveness of influencers' sponsored content on social media platforms.

Second, in exploring the success factors that contribute to the popularity of sponsored posts, this study integrated multiple dimensions, including influencer characteristics, post text, and post image. This approach addresses the limitations of prior research, which often focused on a limited set of factors and lacked a holistic perspective (Mahdikhani, 2022; Purba et al., 2021). Audience evaluations and responses to a post stem from the combined effect of all its elements, which together create a stimulus. Therefore, analyzing specific factors in isolation may not fully capture the underlying mechanism linking post-stimuli to audience responses. Besides, in the context of influencer marketing, beyond textual and visual content, influencer characteristics play a crucial role. Therefore, this study incorporated all three dimensions to more accurately predict and explain the key factors influencing the popularity of sponsored posts, expanding the research perspective in the field of influencer marketing.

Moreover, previous studies on post popularity have primarily focused on the technical aspects, emphasizing the identification of

features or methods that improve prediction accuracy while paying less attention to their practical value for marketing decision-making. As a result, the interpretability of the extracted features (in particular, the image embeddings and word embeddings) has often been overlooked. While such approaches may enhance predictive accuracy, they do not necessarily translate into actionable managerial insights, making it difficult for brand marketers to gain intuitive, easy-to-understand guidance for improving influencer marketing strategies and content planning. In contrast, this study aimed to identify the key success factors of influencers' sponsored posts with a focus on generating business-applicable insights; therefore, the feature extraction methods for text and images were designed with semantic interpretability in mind, primarily utilizing topic modeling. In particular, for images, we not only extracted image embeddings but also the semantic topics after converting images into textual descriptions, thereby successfully quantifying and explaining visual content elements. This approach provides a novel measurement solution for future studies. Additionally, this study employed multiple machine learning models, including Random Forest, XGBoost, Support Vector Machines, Logistic Regression, Deep Neural Networks, and ensemble learning models. These models are thoroughly trained and tested across different feature types, and cross-validation techniques are applied to mitigate model bias, thereby significantly enhancing the credibility and reliability of the results. Furthermore, this study conducted Taguchi experiments to systematically investigate the effects and relative importance of various feature types influencing the popularity of influencer-sponsored posts. Finally, an odds ratio analysis using logistic regression was performed to further examine the impact of specific detailed features, generating actionable insights for optimizing influencer marketing strategies on Instagram.

From a managerial perspective, the results of the Taguchi experiments confirm that, on the Instagram platform, the visual features of images have the most significant impact on the success of influencers' sponsored posts. This finding aligns with the conclusions of Wang et al. (2023) and Antoniadis et al. (2019), which highlight the crucial role of images in post engagement. Additionally, influencer characteristics were identified as the second most important factor in this study, suggesting that, unlike regular posts, the success of sponsored content is influenced by the attributes of the influencer. This aspect has been relatively underexplored in the literature. Brands are recommended to carefully assess an influencer's audience alignment and personal branding when forming partnerships. Furthermore, image topic features also showed a notable impact, helping to supplement the interpretability limitations of visual features, whereas textual features had a relatively weaker influence on sponsored content. This also explains that Instagram is a visually-driven social media platform, where the content strategy should primarily focus on the arrangement of visuals, with text content serving as a supplementary element.

The odds ratio analysis using logistic regression further revealed that highly interactive influencers with a strong presence across multiple platforms—especially female influencers specializing in lifestyle, fashion, and travel—are more likely to attract audiences. Additionally, content should emphasize visual appeal, avoiding static and non-engaging imagery. Instead, vibrant, dynamic visuals such as group photos, lifestyle scenes, or emotionally resonant storytelling images are more effective. For textual content, incorporating topics related to daily life, entertainment, and personal growth enhances audience resonance rather than relying solely on product descriptions or overtly promotional messages. Moreover, sponsored content should avoid excessive commercialization; brand messages are recommended to be naturally integrated into influencers' daily narratives, using authentic experiences and storytelling techniques to build audience trust and engagement.

This study identified key success factors in influencer marketing, enabling brands to select suitable influencers effectively and discern which visual and thematic elements in current collaborations are more likely to capture user interest. When planning marketing content, brands can proactively incorporate these valuable elements into post imagery and captions. Furthermore, in the era of big data, brands should establish systematic data collection and analysis mechanisms, leveraging data-driven insights rather than relying solely on subjective judgment. Predictive modeling can help brands estimate the performance of sponsored posts, thereby optimizing marketing decisions. These findings hold significant practical value for brands, advertising agencies, and influencer marketing strategists.

Finally, while this study highlighted popular topics that enhance content engagement, market trends evolve rapidly. Both brands and influencers must continuously monitor market dynamics and promptly adapt to emerging content trends and shifting user preferences. By leveraging data analytics to uncover hidden key factors, brands can achieve precision marketing, enhance their competitive edge, and maximize the return on marketing investments.

6.2. Limitations and future research

This study offers valuable insights into influencer marketing on Instagram; however, several limitations should be acknowledged, providing opportunities for future research.

First, This study focused solely on Instagram, while influencer marketing often spans multiple platforms such as Facebook, Twitter, and YouTube. Although some cross-platform numerical indicators were considered, the unique characteristics and user demographics of different platforms may lead to varying marketing dynamics. Future research can incorporate data from multiple platforms to provide a more holistic understanding of influencer marketing strategies. Second, prior research suggested that the majority of post engagement (e.g., views and likes) occurs within the first 24 h after publication. However, the total accumulation of engagement over time remains influenced by the duration a post remains visible and relevant. This study did not explicitly account for the long-term temporal effects on engagement, which could be explored in future work. Third, The study did not examine the characteristics of sponsored products, such as category, price, and design. These factors are likely to influence influencer marketing effectiveness. Future research could analyze how different product attributes impact the optimal strategies for influencer endorsements. Fourth, this study primarily focused on post engagement metrics, while user-generated comments, sentiment, and qualitative feedback were not included. Incorporating sentiment analysis and textual evaluations of influencer posts could enhance understanding of audience perception, allowing for a more comprehensive assessment that integrates brand, product, influencer, and consumer perspectives. Finally, the study primarily analyzed text and images separately. Future research could leverage multimodal machine learning models that integrate various data types, such as text, images, and videos, to enhance predictive accuracy and offer richer insights into influencer marketing effectiveness.

CRediT authorship contribution statement

Yu-Hsiang Hsiao: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yi-Yi Lin:** Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yu-Hsiang Hsiao reports financial support was provided by National Science and Technology Council, Taiwan. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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